Medical Emergency Portal – Project Report

1. Introduction

Emergency medical services rely on rapid access to accurate information, clear communication channels, and timely decision-making. In many regions, especially low-resource settings, patients and caregivers must search through fragmented phone directories, paper forms, or scattered web pages to locate the nearest hospital, ambulance service, or first-aid resource. This leads to delays, confusion, and, in worst-case scenarios, preventable loss of life.

Medical Emergency Portal is a full-stack web application that consolidates essential emergency-care information into a single, easy-to-navigate interface. The portal lets users quickly find nearby hospitals, view real-time ambulance availability, read first-aid guides, and contact emergency hotlines—all from a responsive, mobile-friendly site.

2. Problem Statement and Objectives

2.1 Problem Statement

- **Fragmented information** – Hospital locations, contact numbers, and ambulance status are often stored in separate databases or static pages, making it hard for users to locate the right service quickly.
- **Lack of real-time updates** – Users cannot see whether an ambulance is currently available or if a hospital's emergency department is at capacity.
- **Limited accessibility** – Existing solutions are not optimized for low-bandwidth or mobile devices, reducing usability for people in urgent need.

2.2 Project Objectives

1. Provide a central web portal where users can search for hospitals, ambulance services, and first-aid resources by location or specialty.
2. Display real-time availability of ambulances and emergency department capacity (simulated for the demo).
3. Offer concise first-aid guides for common emergencies (e.g., cardiac arrest, choking, bleeding).
4. Enable one-click emergency calls that open the device's dialer with the appropriate hotline number.
5. Deliver a responsive UI that works smoothly on desktops, tablets, and smartphones.

3. Scope of the Project

3.1 Application Scope

- **User-friendly search** – Autocomplete location search with map integration (Leaflet/Mapbox).
- **Hospital directory** – List of hospitals with address, contact number, specialty tags, and a "View on map" button.
- **Ambulance tracker** – Dashboard showing available ambulances, estimated response times, and a "Request ambulance" action (demo only).
- **First-aid library** – Card-style cards with step-by-step instructions and illustrative icons.
- **Emergency call shortcuts** – Clickable phone-link buttons that trigger the device's dialer.

3.2 Technical Boundaries

- The portal is a **demo-level application**; it does not integrate with real-world dispatch systems or hospital EHRs.
- **Authentication** is limited to an admin view for managing hospital/ambulance data; public users have read-only access.
- No **payment processing**, **patient records**, or **advanced analytics** are included.

4. Technologies and Tools Used

Layer	Technology	Purpose
Frontend	**Next.js (React 19)**, **Tailwind CSS**, **Radix UI**, **Framer Motion**	Modern, server-rendered React UI with responsive design and smooth animations
State Management	**Zustand**, **React Hook Form**, **Zod**	Simple global state, form handling, and validation
Backend API	**Express.js**, **Node.js**	REST endpoints for hospitals, ambulances, and first-aid content
Database	**MongoDB (Memory Server for dev)**	Stores hospital/ambulance records and first-aid articles
Authentication	**JSON Web Tokens (JWT)**	Secures admin routes
Styling & Icons	**Tailwind-merge**, **Lucide-react**, **Orchids-visual-edits**	Consistent design system and iconography
Testing	**Jest**, **React Testing Library**	Unit and integration tests for UI components and API routes
Deployment	**Vercel** (or any Node-compatible host)	Simple, serverless deployment for the Next.js app

5. Workflow, Testing & Conclusion

5.1 User Workflow

1. **Search** – The user enters a city or zip code; matching hospitals appear with distance and specialty tags.
2. **View Details** – Clicking a hospital shows address, phone number, and a "Call now" button.
3. **Check Ambulance** – The ambulance dashboard lists available units; the user can tap "Request" (demo).
4. **Read First-Aid** – Selecting a condition opens a step-by-step guide with illustrations.

5.2 Testing Strategy

- **Component tests** for search bar, hospital cards, and first-aid modals using React Testing Library.
- **API tests** for CRUD operations on hospitals/ambulances with Jest and Supertest.
- **End-to-end smoke test** using Playwright to verify that the home page loads, a search returns results, and the "Call now" link is correctly formatted.

5.3 Conclusion

The **Medical Emergency Portal** demonstrates how a modern full-stack web stack can be leveraged to create a fast, accessible, and visually appealing emergency-care information hub. By centralizing critical data and providing real-time (simulated) availability, the portal reduces the time it takes for users to locate help, which can be lifesaving in urgent situations. The project showcases clean architecture, responsive design, and a solid testing foundation, making it a strong foundation for future extensions such as real-time dispatch integration, multilingual support, or offline capabilities.

Feel free to let me know if you'd like any section expanded, additional screenshots, or further customizations.